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Working Group XI - High-redshift Galaxies

Working Group XI - Blog

Untitled Post This is the first blog post for the Working Group XI – High–redshift Galaxies.

Posted Oct 4, 2012, 12:57 AM by Ji-hoon Kim

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– Science Question: Cold flows, clumpiness, kinematics, Lyman–limit systems, etc.

– Description: This working group will be focus on scientific questions related to high–redshift ($z > 1-2$) gas–rich, galaxies at different galaxy mass scales. This rather incomplete list includes: 1) Gas kinematics. Particular attention would be given to the understanding of the origin of rotation–dominated galaxies and dispersion–dominated galaxies. 2) Violent disc instability (and turbulence) in clumpy, gas–rich galaxies at different galaxy mass scales. 3) Cold flows and their importance for gas accretion into galaxies. 4) Detectability of cold flows as Lyman–limit systems and/or metal–line absorbers. [\[authored by Daniel Ceverino\]](#)

In addition to the topics listed above, we could also examine the DLA distribution (more generally H_I gas including LLS), stellar mass/luminosity distribution, and SF histories in these high– z galaxies. The galaxy stellar mass/luminosity function would be difficult due to limited volume and sample. The UNLV group is particularly interested in $z=3-10$ galaxies.

Some of the science questions raised in the Outflow –WG are very relevant to this group as well; for example,

-- What are the roles of galactic outflows on the regulation of star formation, on galaxy morphology and other galaxy properties?

-- If feedback properties vary with galaxy mass and SFR, can we learn something from zoom–in simulations about the luminosity function?

I think these are topics that this WG is also interested in very much. Simulations can definitely be common to the two groups. [\[by Ken Nagamine, Jason Jaacks, Robert Thompson\]](#)